



Koneru Lakshmaiah Education Foundation

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OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026–27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 11

Team No.: 8

Project Title: Linux Dynamic Memory Allocation and Memory Monitoring System

Team Members

Roll Number	Student Name	Signature
2520090047	M Harshika	
2520090013	B Neha	
2520090044	S Chakrika Redddy	

1. Abstract

Dynamic memory management is an important function of an Operating System that allows processes to allocate and release memory during program execution. Improper memory allocation can lead to memory leaks, fragmentation, inefficient memory utilization, and reduced system performance. This project proposes a Linux-based Dynamic Memory Allocation and Memory Monitoring System to demonstrate and monitor how memory is allocated, used, and released by processes.

The system will be implemented using C/C++ in a Linux environment and will use dynamic memory allocation mechanisms such as malloc(), calloc(), realloc(), and free(). Linux system calls and APIs such as mmap(), munmap(), and process memory information from the /proc filesystem will be used to observe and analyze memory usage. The system will allow users to perform memory allocation and deallocation operations while monitoring parameters such as allocated memory, available memory, process memory usage, and memory utilization.

The proposed system aims to provide a practical understanding of memory management concepts in Operating Systems and demonstrate the interaction between application programs and Linux memory management mechanisms. The expected outcome is a simple monitoring interface that displays memory allocation activities and helps identify inefficient memory usage, memory leaks, and changes in memory consumption during program execution.

2. Problem Statement

Efficient memory management is essential for maintaining the performance and stability of a Linux system. Processes continuously allocate and release memory during execution, and improper memory management can result in memory leaks, memory fragmentation, excessive memory consumption, and inefficient utilization of available resources. However, it can be difficult to understand and observe these memory allocation activities at the process level.

The proposed project addresses this problem by developing a Linux-based system that demonstrates dynamic memory allocation and provides real-time monitoring of memory usage. The system will allow users to allocate, reallocate, and release memory while observing changes in memory consumption. It will use Linux memory management mechanisms and system programming interfaces to monitor process-level memory usage and provide useful information about memory utilization.

The project is relevant to Operating Systems and Systems Programming because it demonstrates how applications interact with the operating system for memory management and how Linux manages memory resources for processes.

3. Objectives

1. To implement dynamic memory allocation and deallocation using Linux and C/C++ programming.
2. To demonstrate the use of memory management functions such as `malloc()`, `calloc()`, `realloc()`, and `free()`.
3. To monitor and display the memory usage of processes during program execution.
4. To identify and analyze inefficient memory usage, memory leaks, and changes in memory consumption.
5. To provide a practical understanding of Linux memory management mechanisms and their role in efficient system resource utilization.

4. Proposed Methodology

The project will be developed in a Linux environment using C/C++. The system will be designed with two main functionalities: dynamic memory allocation and memory monitoring.

First, the user will be provided with options to allocate, reallocate, and deallocate memory dynamically using functions such as `malloc()`, `calloc()`, `realloc()`, and `free()`. The system will record the memory allocation and deallocation activities performed during program execution.

Next, the system will monitor memory usage by accessing Linux process and system memory information through the `/proc` filesystem and relevant Linux APIs. The collected information will be processed and displayed to the user, including allocated memory, process memory usage, and available memory.

The overall flow will be: **User Input** → **Memory Allocation/Deallocation** → **Memory Usage Monitoring** → **Data Processing** → **Display of Memory Statistics**. The system will help users observe changes in memory consumption and understand the practical working of Linux memory management.

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API / System Call	Purpose in the Project
Dynamic Memory Allocation	To allocate and release memory dynamically during program execution.
malloc() / calloc()	To allocate memory dynamically for processes.
realloc() / free()	To resize allocated memory and release it when no longer required.
mmap() / munmap()	To demonstrate Linux memory mapping and memory region management.
/proc Filesystem	To monitor and obtain process and system memory usage information.

6. Individual Contribution

Roll Number	Student Name	Individual Responsibility
2520090047	M Harshika	Implement memory monitoring using the /proc filesystem and mmap()/munmap() and collect process memory statistics.
2520090013	B Neha	Implement dynamic memory allocation and deallocation using malloc(), calloc(), realloc(), and free().
2520090044	S Chakrika Reddy	Develop the user interface/menu, integrate all modules, test the system, and handle output visualization and documentation.

7. Tools / Platforms / Software Used

Tool / Platform / Software	Purpose
Linux / Ubuntu	Operating system environment for developing and executing the project.
C / C++	To implement dynamic memory allocation, deallocation, and memory monitoring.
GCC Compiler	To compile and execute the C programs in Linux.
Gedit / VS Code	To write and edit the source code.
Terminal	To compile, execute, and test the programs and monitor system information.

8. Expected Outcome

The project is expected to provide a Linux-based system that demonstrates dynamic memory allocation and deallocation using malloc(), calloc(), realloc(), and free(). It will display memory usage information of the system and processes and show how memory consumption changes during program execution. The system will help identify inefficient memory usage and possible memory leaks while providing a practical understanding of Linux memory management and memory monitoring. The final output will be a simple and user-friendly interface through which users can perform memory operations and observe the corresponding memory usage statistics.

9. GitHub Repository

GitHub Repository Link: <https://github.com/Chakrika-Reddy/OS-Project>

Repository Created:

☐ Yes ☐ No

Faculty Name and Signature

Faculty Name: D Alekhya

Signature: _____

Remarks: _____

Faculty Use

Project Approved: _____

☐ Yes ☐ No

Date: _____

Faculty Signature: _____